

Ans. to the ques. no. 1

(a) Let $0^i 1^j 0^k$; $i+k \geq 0$

Let, $w = 0^p 1^p 0^1$

$w \in L$

$[1 \leq p]$

$0^\alpha 0^\beta 0^{p-\alpha-\beta} 1^p 0^1$

Pumping i times,

$0^\alpha 1^i \beta 0^{p-\alpha-\beta} 1^p 0^1$

$\alpha = \alpha + i\beta + p - \alpha - \beta$

$= p + i\beta - \beta$

Here,

$p + i\beta - \beta + 1 \geq p$

or, $-1 + p + \beta(i-1) + 1 \geq p$

or, $p + \beta(i-1) \geq p$

or, $\beta(i-1) \geq 0$

or, $i-1 \geq 0$

or, $i \geq 1$

Here, $i \geq 0$,

If $i = 0$,

$$p + 0 - \beta + 1 > p$$

$$\text{or, } p - \beta + 1 > p$$

$$\text{or, } p - 1 + 1 > p$$

$$\text{or, } p > p$$

$$[\beta \geq 1]$$

Here, $p - \beta$ cannot be $> p$. Thus this is non-regular language

(b) Let, $a^p b^1 a^{p+2}$

$$w \in L$$

$$a^\alpha a^\beta a^{p-\alpha-\beta} b^1 a^{p+2}$$

Pumping i times,

$$a^\alpha a^{i\beta} a^{p-\alpha-\beta} b^1 a^{p+2}$$

Here,

$$\alpha + i\beta + p - \alpha - \beta$$

$$= p + i\beta - \beta$$

Here,

$$p + i\beta - \beta + 1 < p + 2$$

$$\text{or, } p + i\beta - \beta + 2 \leq p + 2$$

$$\text{or, } i\beta - \beta \leq 0$$

$$\text{or, } \beta(i-1) \leq 0$$

$$\text{or, } i-1 \leq 0$$

$$\text{or, } i \leq 1$$

Here,

$$i \geq 0. \text{ if } i=2,$$

$$p + 2\beta - \beta + 2 \leq p + 2$$

$$\text{or, } p + 2 + \beta \leq p + 2$$

Since $\beta \leq 0$ but $\beta \geq 1$ cannot be possible, so the language

is ^{non-}regular.

(c) Let $0^p!$
 $\omega \in L$

Here,

$$0^\alpha 0^\beta 0^{p! - \alpha - \beta}$$

Pumping i -times,

$$0^\alpha 0^{i\beta} 0^{p! - \alpha - \beta}$$

$$= 0^{\alpha + i\beta + p! - \alpha - \beta}$$

$$= p! + i\beta - \beta$$

If $i=2$,

$$p! + 2\beta - \beta$$

$$= p! + \beta$$

$$[\beta \leq 0]$$

$$\text{So, } p! < p! + \beta$$

$$[\beta \leq p]$$

$$\text{or, } p! + \beta \leq p! + p$$

$$\text{or, } p! + p \leq (p+1)!$$

$$\text{or, } p! + p < p! (p+1)$$

But the highest value of β is p , meaning $p! + \beta$ is not possible. Thus it is non-regular language.

(d) Let, $a^p b^{p+p!}$

$w \in L$

Here,

$$a^\alpha a^\beta a^{p-\alpha-\beta} b^{p+p!}$$

Pumping i -times,

$$a^\alpha a^{i\beta} a^{p-\alpha-\beta} b^{p+p!}$$

$$\Rightarrow \alpha + i\beta + p - \alpha - \beta$$

$$\Rightarrow p + i\beta - \beta$$

$$\Rightarrow p + \beta(i-1) \neq p + p!$$

$$\Rightarrow \beta(i-1) \neq p!$$

$$\Rightarrow i-1 \neq \frac{p!}{\beta}$$

$$\therefore i \neq \frac{p!}{\beta} + 1$$

The value condition will not work if the value of $i = \frac{p!}{\beta} + 1$.

Thus it is a non-regular language.

(e) Let, $0^p 10^p$

Here,

$$0^\alpha 0^\beta 0^{p-\alpha-\beta} 0^p$$

Pumping i -times,

$$\begin{aligned} & 0^\alpha 0^i \beta 0^{p-\alpha-\beta} 0^p \\ &= p + i\beta - \beta \\ &= p + \beta(i-1) = p \end{aligned}$$

Here, $i \geq 0$, $\beta \geq 1$. $p + \beta i \neq p$ which makes it a non-regular language.

Ans. to the ques. no. 2

(a) $S \rightarrow S' | S''$

$$S' \rightarrow a S' b | a A$$

$$S'' \rightarrow a S'' b | b B b$$

$$A \rightarrow a A | \epsilon$$

$$B \rightarrow B b | \epsilon$$

(b) $S \rightarrow a a A$

$$A \rightarrow a a a A b | \epsilon$$

(c) $S \rightarrow S' | S''$

$$S' \rightarrow a A_1 b B_1 c$$

$$S'' \rightarrow a A_2 b B_2 c$$

$$A_1 \rightarrow a A_1 b | \epsilon$$

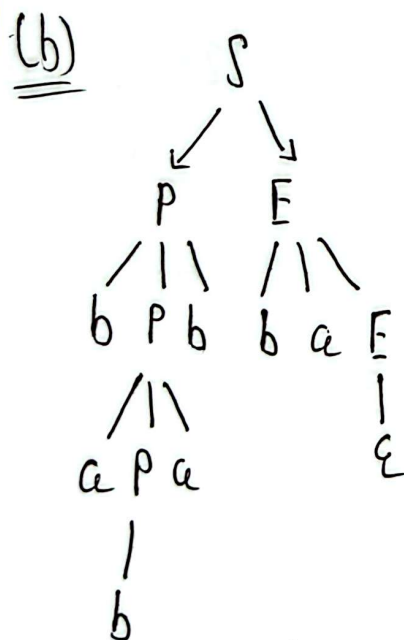
$$A_2 \rightarrow a A_2 | \epsilon$$

$$B_1 \rightarrow B_1 c | \epsilon$$

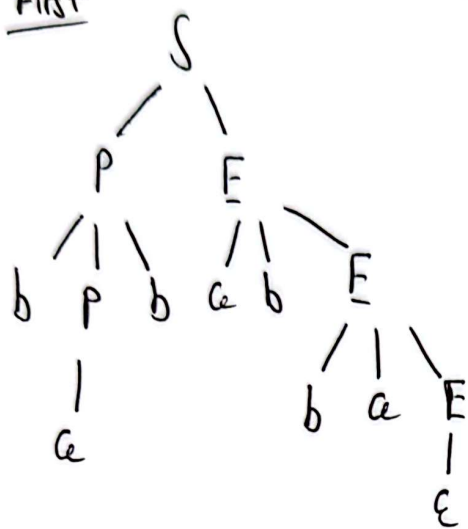
$$B_2 \rightarrow b B_2 c | \epsilon$$

Ans. to the ques. no. 3

(a) $S \rightarrow PE$
 $\xrightarrow{L} bPbE$
 $\xrightarrow{L} baPabE$
 $\xrightarrow{L} bababE$
 $\xrightarrow{L} bababbaE$
 $\xrightarrow{L} bababba$

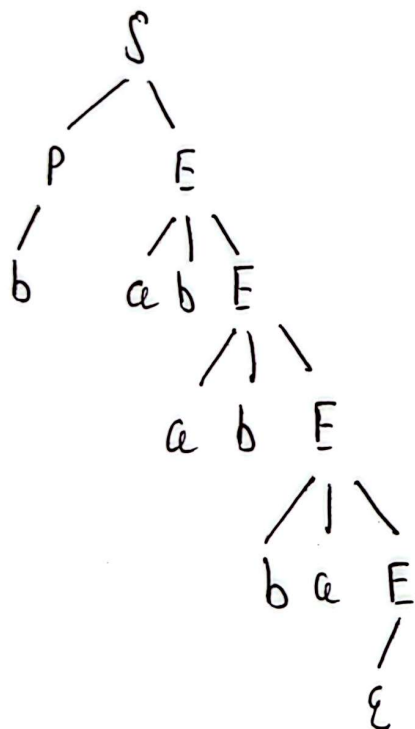


(c) First



$\epsilon b a \epsilon \leftarrow$
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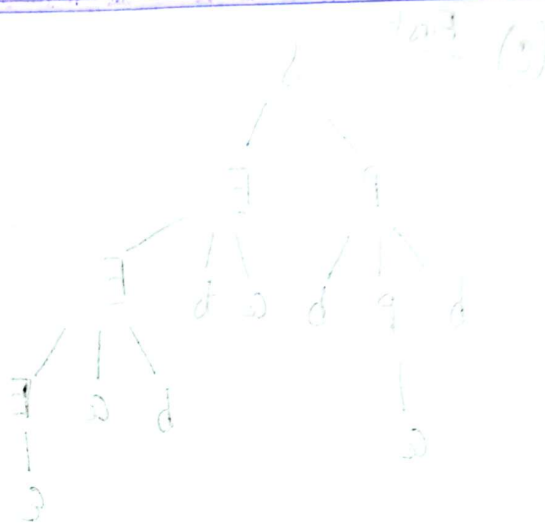
Second



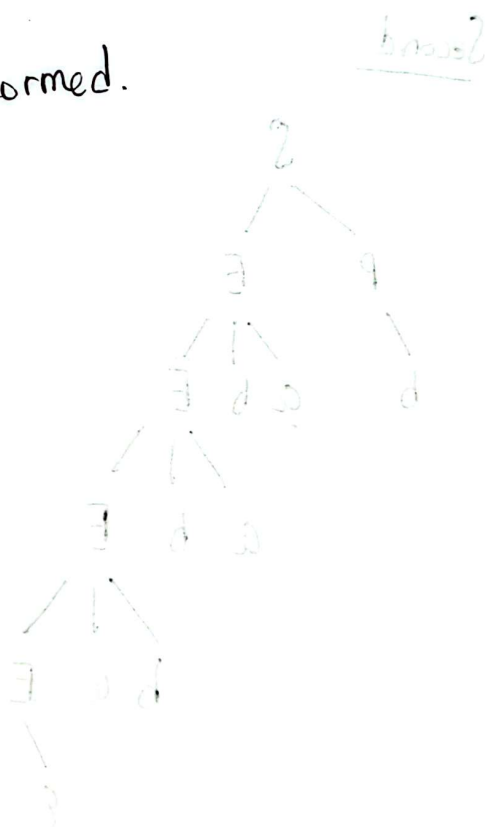
Only one parse tree can be formed

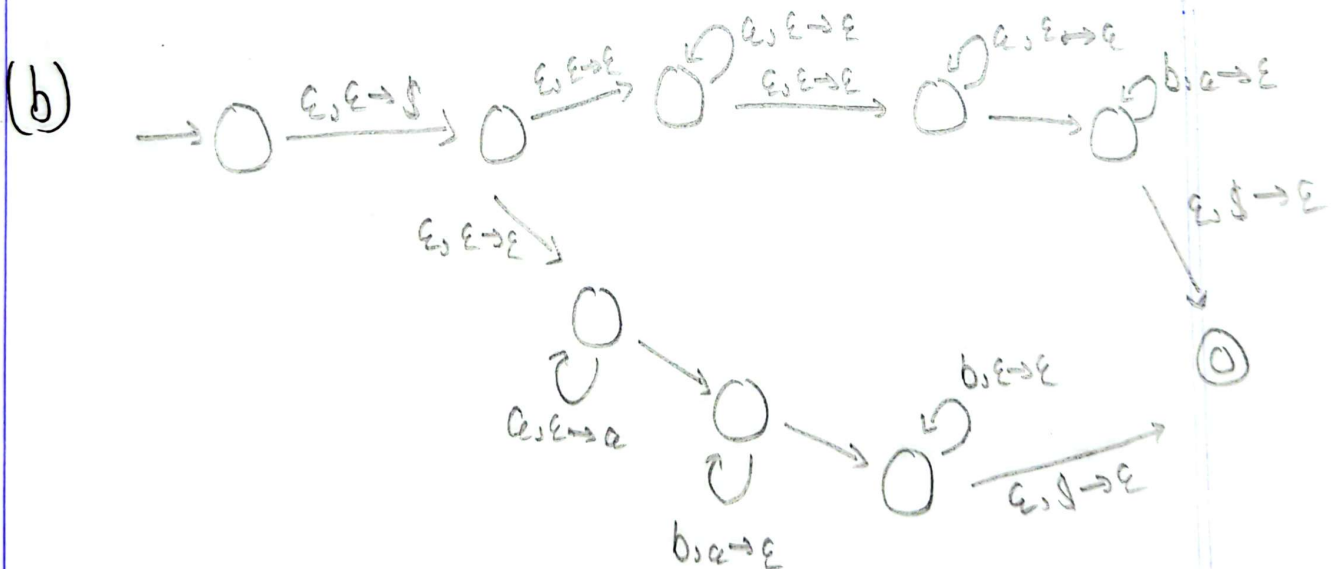
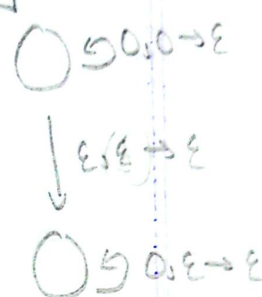
$\therefore$ The two parse trees are ~~am~~ different and thus ambiguous.

(d) $S \rightarrow PE$
 $\rightarrow aE$
 $\rightarrow aabE$
 $\rightarrow aababE$
 $\rightarrow aabababE$
 $\rightarrow aabababab$

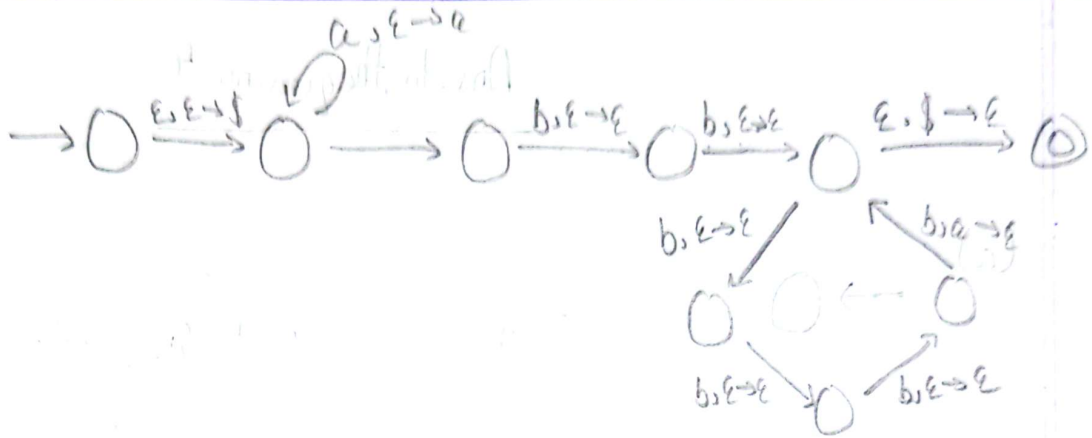


Only one parse tree can be formed.

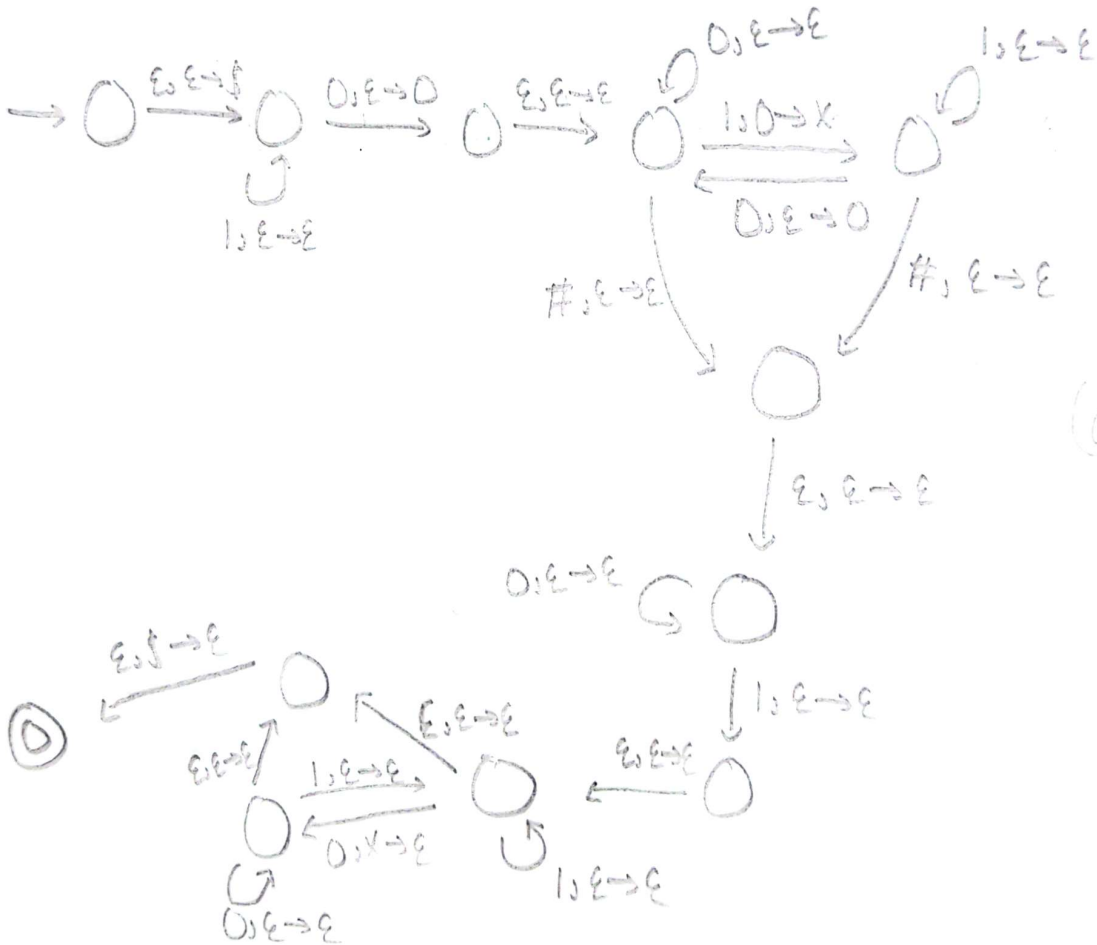




(c)

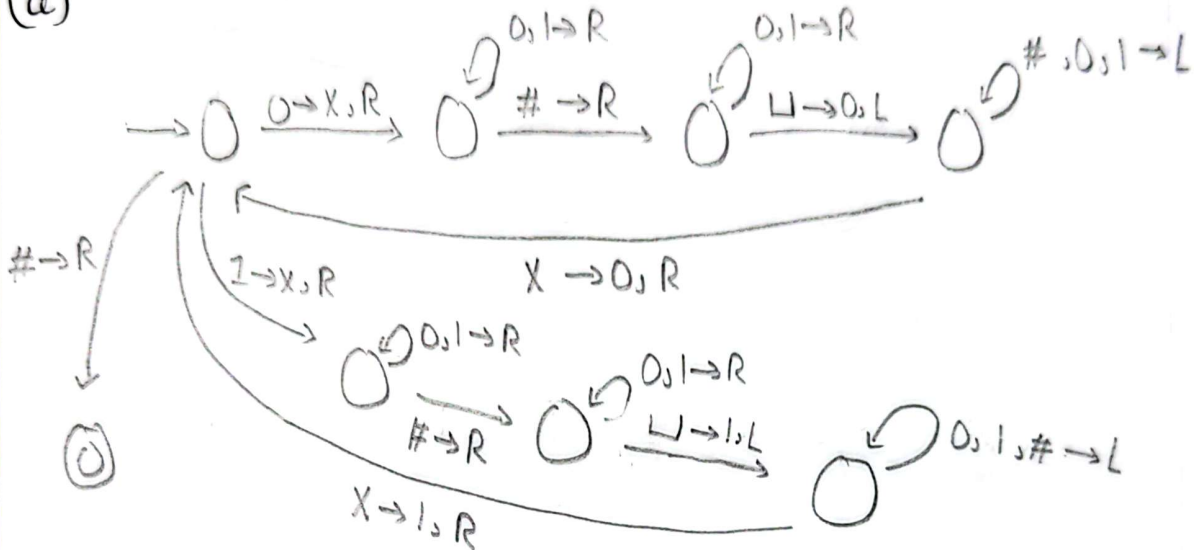


(d)

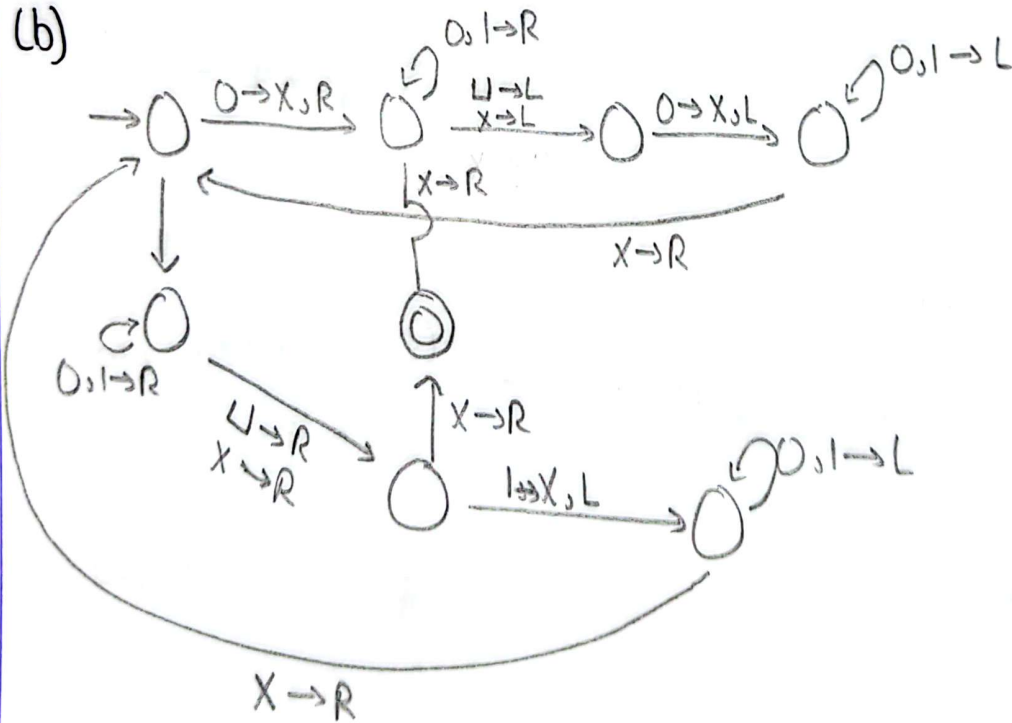


Ans. to the ques. no. 5

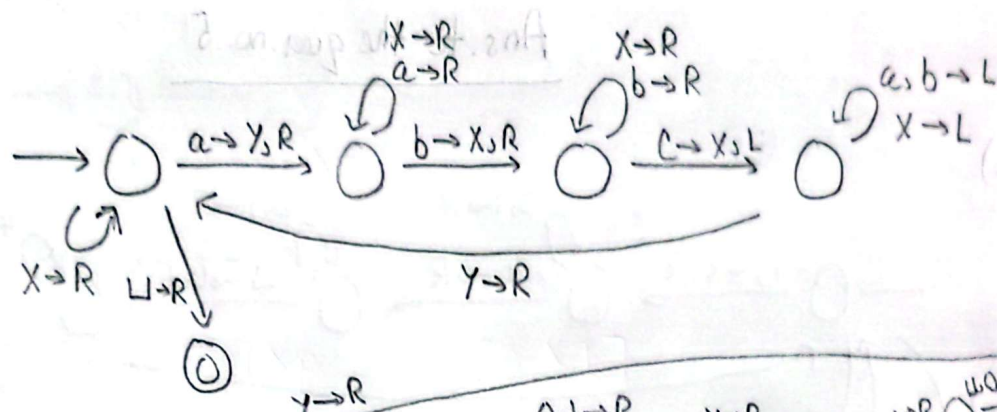
(a)



(b)



(c)



(d)

